Practical No:4.1

Aim: To design a simple Android application using Linear Layout to display student information and understand vertical and horizontal orientation.

Problem Statement:
Design an Android application that displays student details such as Name, Roll Number, Course, and College Name using Linear Layout. Arrange the views vertically and horizontally to understand orientation.

Step 1: Create New Project

Open **Android Studio**

Click **New Project**

Select **Empty Views Activity**

Click **Next**

Fill details:

Name: StudentProfileLinearLayout

Package: com.example.studentprofile

Language: **Java**

Click **Finish**

Step 2: Open Layout File

app > res > layout > activity_main.xml

Switch to **Code view** (XML only)

Source Code:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">
    <!-- Title -->
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Student Profile"
        android:textSize="22sp"
        android:textStyle="bold"
        android:layout_gravity="center"
        android:paddingBottom="20dp" />
    <!-- Name Row -->
```

```
<LinearLayout
   android:layout_width="match_parent"
   android:layout_height="wrap_content"
   android:orientation="horizontal">
   <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Name: "
      android:textStyle="bold" />
   <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Nikhil" />
</LinearLayout>
<!-- Roll Number Row -->
<LinearLayout
   android:layout_width="match_parent"
   android:layout_height="wrap_content"
   android:orientation="horizontal"
   android:paddingTop="10dp">
   <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Roll No: "
      android:textStyle="bold" />
   <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="21MCA001" />
</LinearLayout>
<!-- Course Row -->
<LinearLayout
   android:layout_width="match_parent"
   android:layout_height="wrap_content"
   android:orientation="horizontal"
   android:paddingTop="10dp">
   <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Course: "
      android:textStyle="bold" />
   <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="BSc IT" />
</LinearLayout>
<!-- College Row -->
<LinearLayout
```

```
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal"
        android:paddingTop="10dp">
        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="College: "
            android:textStyle="bold" />
        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="ABC Institute of Technology" />
    </LinearLayout>
</LinearLayout>
```

Step 4: Java File (No Code Needed)

Open:

MainActivity.java

No changes required because:
This is **static UI only**, no button or logic.

Step 5: Run the App

Practical 4.2

Aim: To design a simple login screen using Relative Layout and understand positioning of views relative to parent and other views.

Problem Statement

Create an Android application to design a Login Screen using Relative Layout with Username, Password, and Login Button positioned relative to each other.

Project Name: LoginRelativeLayout

activity_main.xml: Source Code

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="20dp">
    <!-- Title -->
    <TextView
        android:id="@+id/txtTitle"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Login"
        android:textSize="24sp"
        android:textStyle="bold"
        android:layout_centerHorizontal="true" />
    <!-- Username -->
    <EditText
        android:id="@+id/edtUsername"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Username"
        android:layout_below="@id/txtTitle"
        android:layout_marginTop="20dp" />
    <!-- Password -->
    <EditText
        android:id="@+id/edtPassword"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Password"
        android:inputType="textPassword"
        android:layout_below="@id/edtUsername"
        android:layout_marginTop="10dp" />
    <!-- Login Button -->
    <Button
        android:id="@+id/btnLogin"
        android:layout_width="wrap_content"
```

```
        android:layout_height="wrap_content"
        android:text="Login"
        android:layout_below="@id/edtPassword"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="20dp" />
</RelativeLayout>
```

MainActivity.java

```
package com.example.loginrelativelayout;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

Run

Practical 4.3

Problem Statement

Develop an Android application that displays a student mark sheet using **Table Layout**, showing subject names and corresponding marks in rows and columns.

Project Name: MarkSheetTableLayout

Source Code: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<TableLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="16dp"
    android:stretchColumns="1">
    <!-- Title -->
    <TableRow>
        <TextView
            android:layout_span="2"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Student Mark Sheet"
            android:textSize="22sp"
            android:textStyle="bold"
            android:gravity="center" />
    </TableRow>
    <!-- Header Row -->
    <TableRow>
        <TextView
            android:text="Subject"
            android:textStyle="bold" />
        <TextView
            android:text="Marks"
            android:textStyle="bold" />
    </TableRow>
    <!-- Row 1 -->
    <TableRow>
        <TextView android:text="Mathematics" />
        <TextView android:text="85" />
    </TableRow>
    <!-- Row 2 -->
    <TableRow>
        <TextView android:text="Computer Science" />
        <TextView android:text="90" />
    </TableRow>
    <!-- Row 3 -->
    <TableRow>
```

```
        <TextView android:text="Physics" />
        <TextView android:text="78" />
    </TableRow>
    <!-- Row 4 -->
    <TableRow>
        <TextView android:text="English" />
        <TextView android:text="88" />
    </TableRow>
</TableLayout>
```

Run

Prac-4.4

Problem Statement

Develop an Android application that displays a list of contact names entered by the user using **ListView**.

Project Name: ContactListViewApp

activity_main.xml

SourceCode:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Contact List"
        android:textSize="22sp"
        android:textStyle="bold"
        android:layout_gravity="center"
        android:paddingBottom="10dp"/>
    <EditText
        android:id="@+id/edtName"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter contact name"/>
    <Button
        android:id="@+id/btnAdd"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Add"
        android:layout_gravity="center"
```

```
        android:paddingTop="10dp"/>
    <ListView
        android:id="@+id/listView"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:paddingTop="10dp"/>
</LinearLayout>
```

MainActivity.java

SourceCode:

```
package com.example.contactlistviewapp;
import android.os.Bundle;
import android.view.View;
import android.widget.ArrayAdapter;
import android.widget.Button;
import android.widget.EditText;
import android.widget.ListView;
import androidx.appcompat.app.AppCompatActivity;
import java.util.ArrayList;
public class MainActivity extends AppCompatActivity {
    EditText edtName;
    Button btnAdd;
    ListView listView;
    ArrayList<String> contactList;
    ArrayAdapter<String> adapter;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        edtName = findViewById(R.id.edtName);
        btnAdd = findViewById(R.id.btnAdd);
        listView = findViewById(R.id.listView);
        contactList = new ArrayList<>();
        adapter = new ArrayAdapter<>(
                this,
                android.R.layout.simple_list_item_1,
                contactList
        );
        listView.setAdapter(adapter);
        btnAdd.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                String name = edtName.getText().toString();
                if (!name.isEmpty()) {
                    contactList.add(name);
                    adapter.notifyDataSetChanged();
                    edtName.setText("");
```

```
            }
        }
    });
  }
}
```

Click **Run**

Prac-4.5

Problem Statement

Design an Android application using **GridView** that allows the user to **enter items dynamically** and display them in a **grid format** on the screen.

Aim

To create an Android application that uses **GridView with Adapter** to display user-entered items dynamically in a grid arrangement.

Project Name: DynamicGridViewApp

activity_main.xml

source Code:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:orientation="vertical"
  android:padding="16dp">
  <EditText
    android:id="@+id/etItem"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter item" />
  <Button
    android:id="@+id/btnAdd"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Add to Grid" />
  <GridView
    android:id="@+id/gridView"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:numColumns="2"
    android:horizontalSpacing="10dp"
    android:verticalSpacing="10dp" />
</LinearLayout>
```

MainActivity.java

SourceCode:

```
package com.example.dynamicgridviewapp;
import android.os.Bundle;
import android.view.View;
import android.widget.ArrayAdapter;
import android.widget.Button;
import android.widget.EditText;
import android.widget.GridView;
import androidx.appcompat.app.AppCompatActivity;

import java.util.ArrayList;
public class MainActivity extends AppCompatActivity {
    EditText etItem;
    Button btnAdd;
    GridView gridView;
    ArrayList<String> itemList;
    ArrayAdapter<String> adapter;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        etItem = findViewById(R.id.etItem);
        btnAdd = findViewById(R.id.btnAdd);
        gridView = findViewById(R.id.gridView);
        itemList = new ArrayList<>();
        adapter = new ArrayAdapter<>(
                this,
                android.R.layout.simple_list_item_1,
                itemList
        );
        gridView.setAdapter(adapter);
        btnAdd.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String item = etItem.getText().toString();
                if (!item.isEmpty()) {
                    itemList.add(item);
                    adapter.notifyDataSetChanged();
                    etItem.setText("");
                }
            }
        });
    }
}
```

Run